

8

[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 1095

I

Unique Paper Code : 2342013502

Name of the Paper : Theory of Computation

**Name of the Course : B.Sc. (Hons.) Computer
Science**

Semester : V

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. The paper has **two** sections. **Section A** is compulsory.
3. Attempt any **four** questions from **Section B**.
4. Part of the questions to be attempted together.
5. Assume $\Sigma = \{a, b\}$ is the underlying alphabet unless mentioned otherwise.

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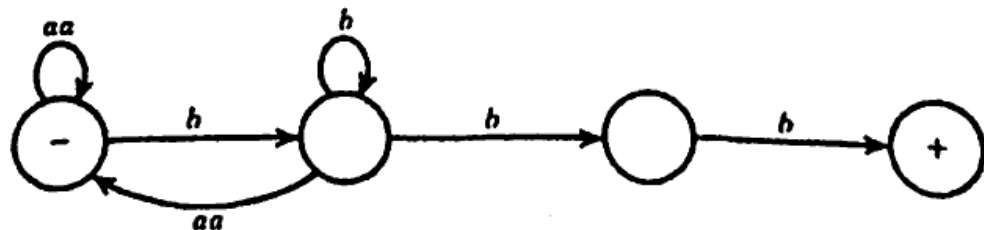
Section A

1. (a) Prove that for all sets S , $(S^+)^* = (S^*)^*$. (3)
- (b) Prove that if x is in PALINDROME then so is x^n for any n . (3)
- (c) Construct a Regular expression for the language consisting of all words with exactly two a's. (3)
- (d) Consider the language S^* , where $S = \{aa, b\}$. How many words of this language have a length of 6? (3)
- (e) Construct a Deterministic Finite Automata that accepts all strings from the language $L = \{\text{all strings with odd number of b's}\}$. (3)
- (f) Determine whether the language $L = \{w \mid w = a^m b^n \text{ and } m, n \geq 1\}$ regular or not. Justify. (3)

- (g) Show that the complement of a context free language may or may not be context free. (4)
- (h) Find the Context Free Language for language $L = \{aa^n b^n \text{ where } n \geq 0\}$. (4)
- (i) Design a Turing machine for the language $a(a+b)^*$. (4)

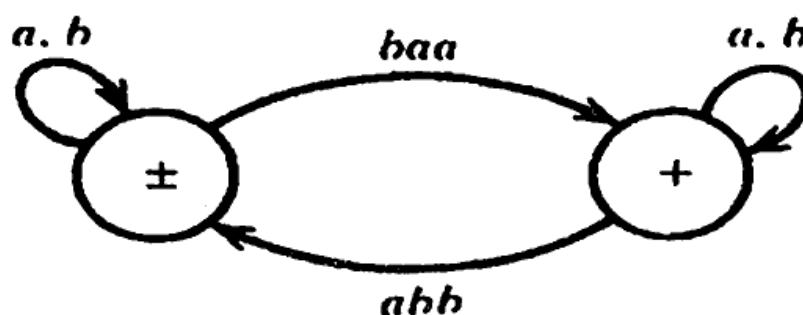
Section B

2. (a) Describe in English language the language accepted by the following automata (2)



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- (b) Construct a regular expression for all words that contain two b's or exactly three b's. (3)
- (c) Build a finite automata that accepts only those words that do not end with ba. (5)
- (d) Convert the following Transition Graph into Regular Expression (5)



3. (a) Consider the Context Free Grammar (CFG):

$$S \rightarrow aX, X \rightarrow aX \mid bX \mid \wedge.$$

What is the language generated by above CFG? (3)

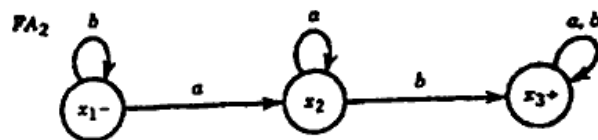
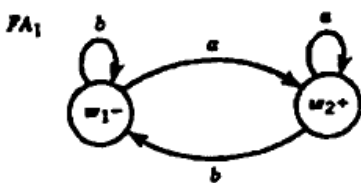
(b) What do you understand by Total language tree?

Explain with an example. (6)

(c) Explain what this machine does. (6)

$$>R \xrightarrow{a \neq \cup} R \xrightarrow{b \neq \cup} R \cup a R \cup b$$

4. (a) For the following Finite Automata, find a finite automaton for $FA_1 \cup FA_2$. (5)



(b) What do you understand by regular language? If

L_1 and L_2 are two regular languages, then so are

$L_1 + L_2$ and $L_1 L_2$. Justify your answer. (5)

P.T.O.

- (c) Convert the given Context Free Grammar into Chomsky Normal Form :

$$S \rightarrow aXX, X \rightarrow aS \mid bS \mid a \quad (5)$$

5. (a) Construct a finite automata for $L_1 \cap L_2$ where $L_1 = (a+b)^*a$ and $L_2 = b(a+b)^*$. (5)

- (b) Using pumping lemma show that the language $L = \{a^{n+1}b^n \text{ where } n = 1\ 2\ 3\ \dots\}$ is non regular. (5)

- (c) Find the Context Free Grammar for

(i) a^*b^*

(ii) $(ab+ba)^*$ (5)

6. (a) Show that the Context Free Grammar,

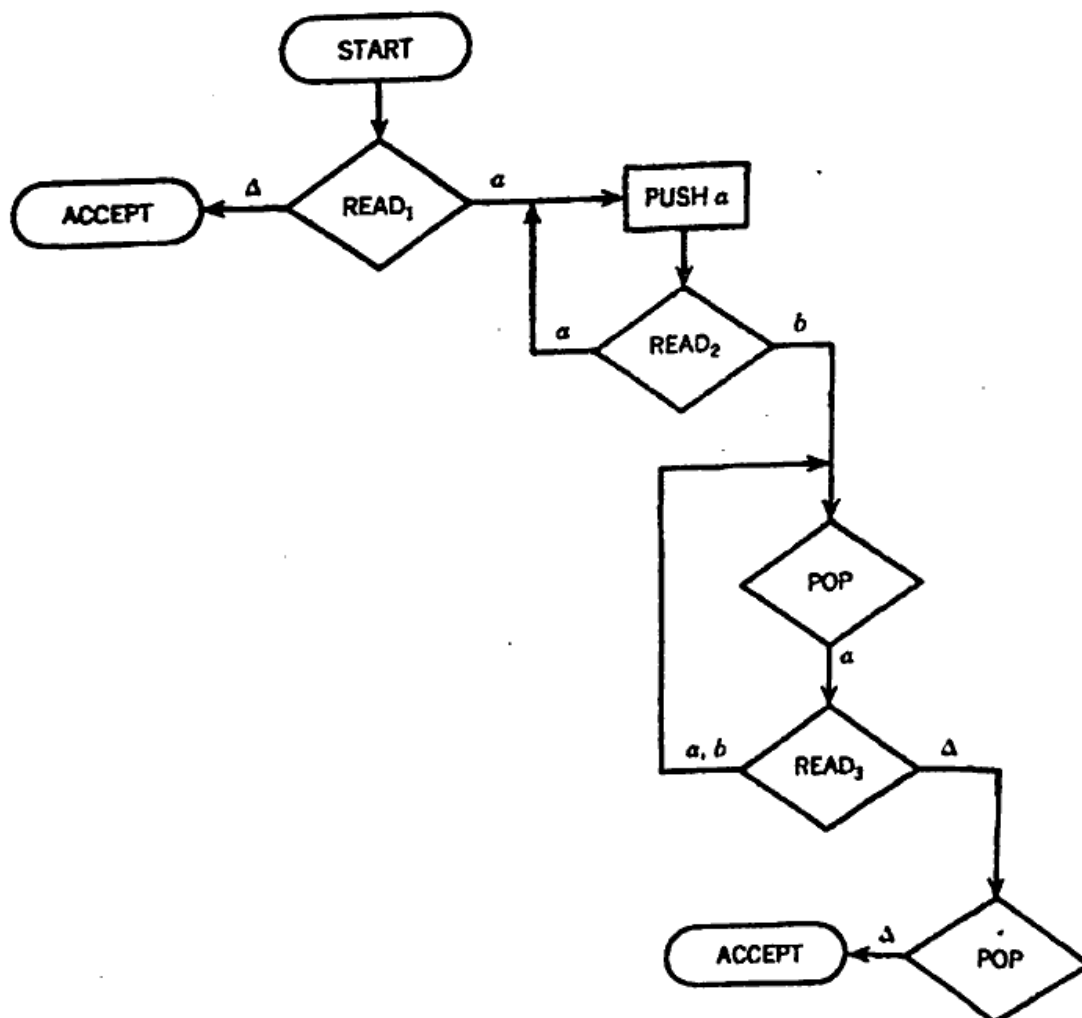
$$S \rightarrow aSb \mid Sb \mid Sa \mid a \text{ is ambiguous.} \quad (5)$$

(b) Design a Push down automata for language $L = \{a^{2n}b^n \text{ where } n = 0, 1, 2, 3, \dots\}$. (5)

(c) Prove that the language $\{a^n b^n a^n b^n a^n \text{ where } n = 1, 2, 3, 4, \dots\}$ is non context free. (5)

7. (a) Explain Church Turing thesis. (3)

(b) Consider the following Push Down Automata : (6)



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